Project Scope Management

Introduction · Plan Scope · Collect Requirements · Define Scope · Create WBS · Validate Scope · Control Scope

What is Project Scope Management?

Scope

All the work involved in creating the products of the project and the processes used to create them.

Deliverable

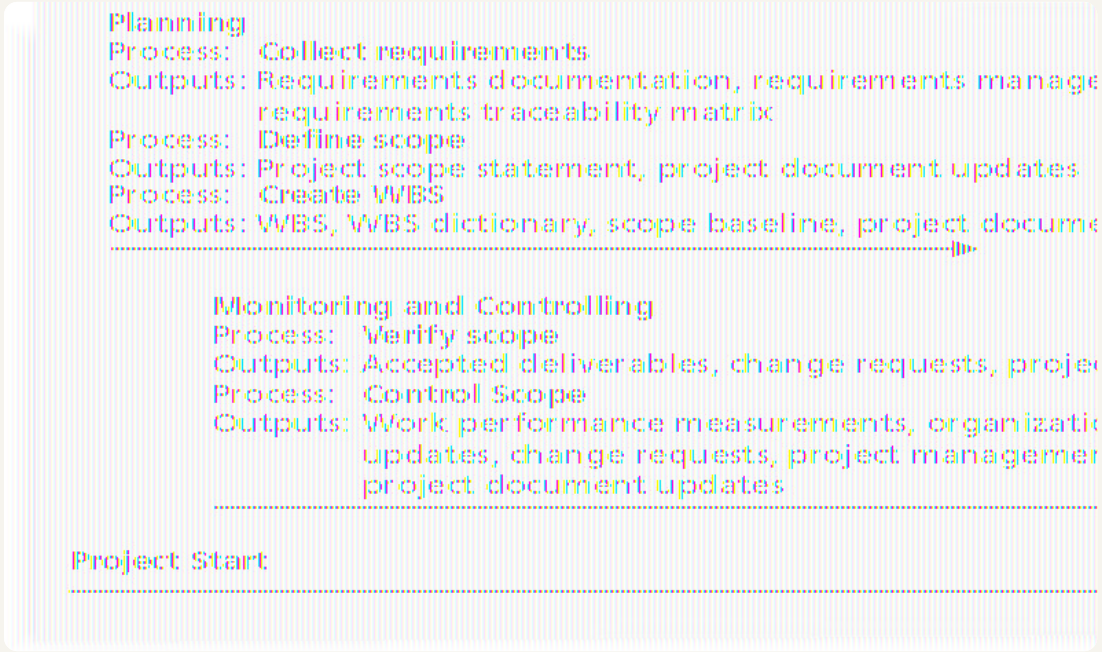
A product produced as part of a project: hardware or software, planning documents, or meeting minutes.

Scope management

The processes for defining and controlling what is or is not included in the project.

Scope Management Processes

- **Collect requirements:** define and document features and functions
- **Define scope:** review charter and requirements, write the scope statement
- **Create WBS:** subdivide deliverables into manageable parts
- **Verify scope:** formal acceptance of deliverables
- **Control scope:** control changes throughout the project



Processes by Process Group

	initiation	planning	execution	monitoring and control	closing
scope management processes		collect requirements define scope create WBS		verify scope control scope	

Collect requirements, define scope and create WBS happen in **planning**. Verify scope and control scope belong to **monitoring and controlling**.

Plan Scope Management

Decide **how** scope and requirements will be defined, documented, validated and controlled.

Inputs

- Project charter
- Project management plan

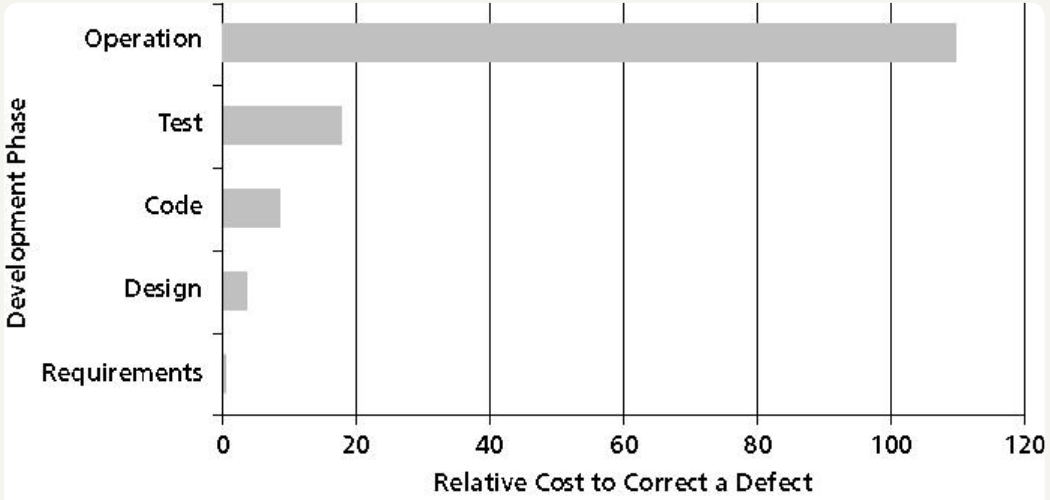
Outputs

- Scope management plan
- Requirements management plan: how requirements will be analyzed, documented and managed

Collect Requirements

A requirement is a condition or capability a system, product or result must have to satisfy a contract, standard or specification.

- Define requirements **iteratively**: they are unclear early on
- Fixing a defect in later phases costs far more than in the requirements phase



Source: Robert B. Grady, "An Economic Release Decision Model: Insights into Software Project Management." *Proceedings of the Applications of Software Measurement Conference* (Orange Park, FL: Software Quality Engineering, 1999), pp.227–239.

Methods for Collecting Requirements

Method	Strength / when to use
Interviews (one-on-one)	Effective but expensive and time consuming
Focus groups, facilitated workshops	Faster and cheaper than individual interviews
Questionnaires and surveys	Efficient if stakeholders answer honestly and thoroughly
Observation	Projects improving work processes and procedures
Prototyping	Software development projects

Software tools can also help collect and manage requirements.

Documenting Requirements

Documents can include text, images, diagrams and video, grouped as functional, service, performance, quality or training requirements. A **requirements traceability matrix (RTM)** lists each requirement, its attributes and status so none is missed.

Req. No.	Name	Category	Source	Status
R32	Laptop memory	Hardware	Project charter and corporate laptop specifications	Complete: laptops ordered have 4 GB of memory

Sample RTM from the textbook.

Define Scope

Project scope is the work needed to create products that meet the requirements.

Good scope definition helps

- Improve accuracy of time, cost and resource estimates
- Define a baseline for performance measurement
- Assign clear work responsibility

Inputs and outputs

- **Inputs:** project charter, requirements documentation
- **Outputs:** project scope statement, project document updates
- Scope becomes clearer as the project progresses

Example: Further Defining Project Scope

TABLE 5-3 Further defining project scope

Project Charter: Upgrades may affect servers . . . (listed under Project Objectives)
Project Scope Statement, Version 1: Servers: If additional servers are required to support this project, they must be compatible with existing servers. If it is more economical to enhance existing servers, a detailed description of enhancements must be submitted to the CIO for approval. See current server specifications provided in Attachment 6. The CEO must approve a detailed plan describing the servers and their location at least two weeks before installation.
Project Scope Statement, Version 2: Servers: This project will require purchasing ten new servers to support Web, network, database, application, and printing functions. Virtualization will be used to maximize efficiency. Detailed descriptions of the servers are provided in a product brochure in Appendix 8 along with a plan describing where they will be located.

Creating the Work Breakdown Structure

What it is

A deliverable-oriented grouping of project work that defines the total scope.

Decomposition: subdividing work into smaller pieces.

Why it matters

A foundation document for planning, managing schedules, costs and resources, and managing project changes.

Work package

A task at the lowest WBS level. It cannot be decomposed into smaller tasks.

Ways to Organize a WBS

By product

Level 2 items are major deliverables of the system.

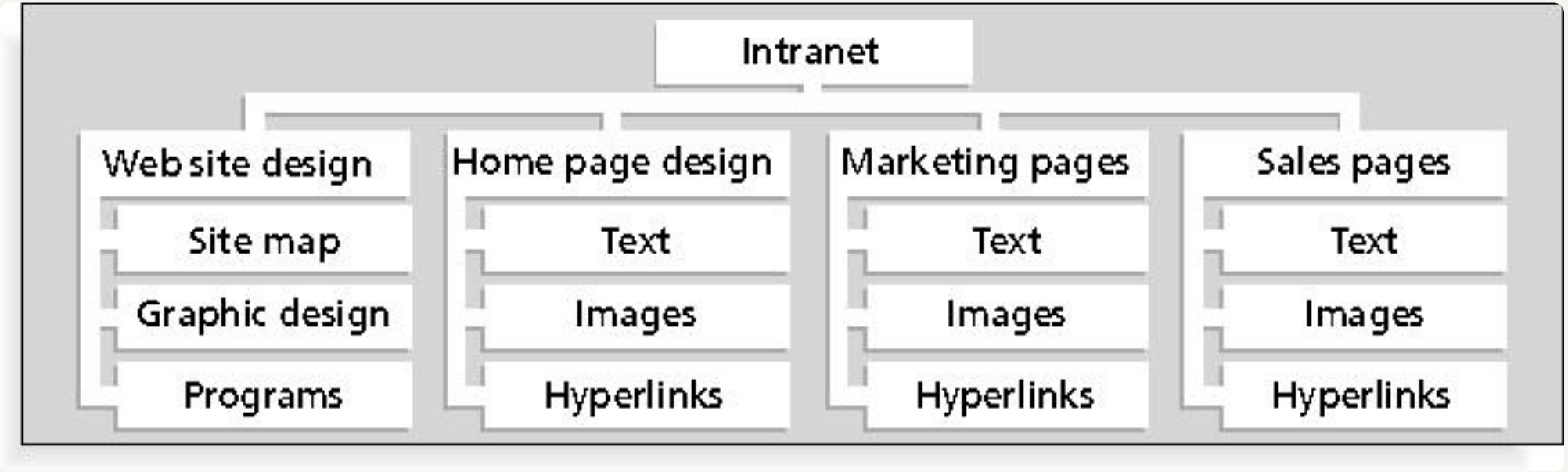
By phase

Level 2 items are project phases such as concept, design, development and roll out.

By process group

Level 2 items are initiating, planning, executing, monitoring and controlling, and closing.

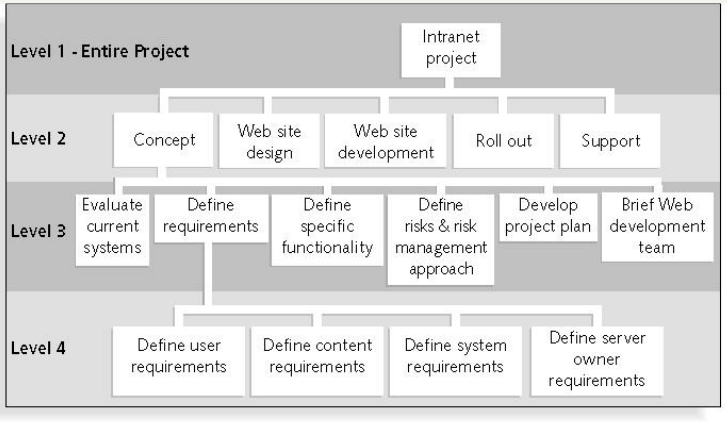
Sample Intranet WBS Organized by Product



Each branch is a deliverable (website design, home page design, marketing pages, sales pages) broken into smaller parts.

Sample Intranet WBS Organized by Phase

Chart form →



Tabular form with Microsoft Project numbering

1.0 Concept
- 1.1 Evaluate current systems
- 1.2 Define requirements
  - 1.2.1 Define user requirements
  - 1.2.2 Define content requirements
  - 1.2.3 Define system requirements
  - 1.2.4 Define server owner requirements
- 1.3 Define specific functionality
- 1.4 Define risks and risk management approach
- 1.5 Develop project plan
- 1.6 Brief Web development team

2.0 Web site design
3.0 Web site development
4.0 Roll out
5.0 Support

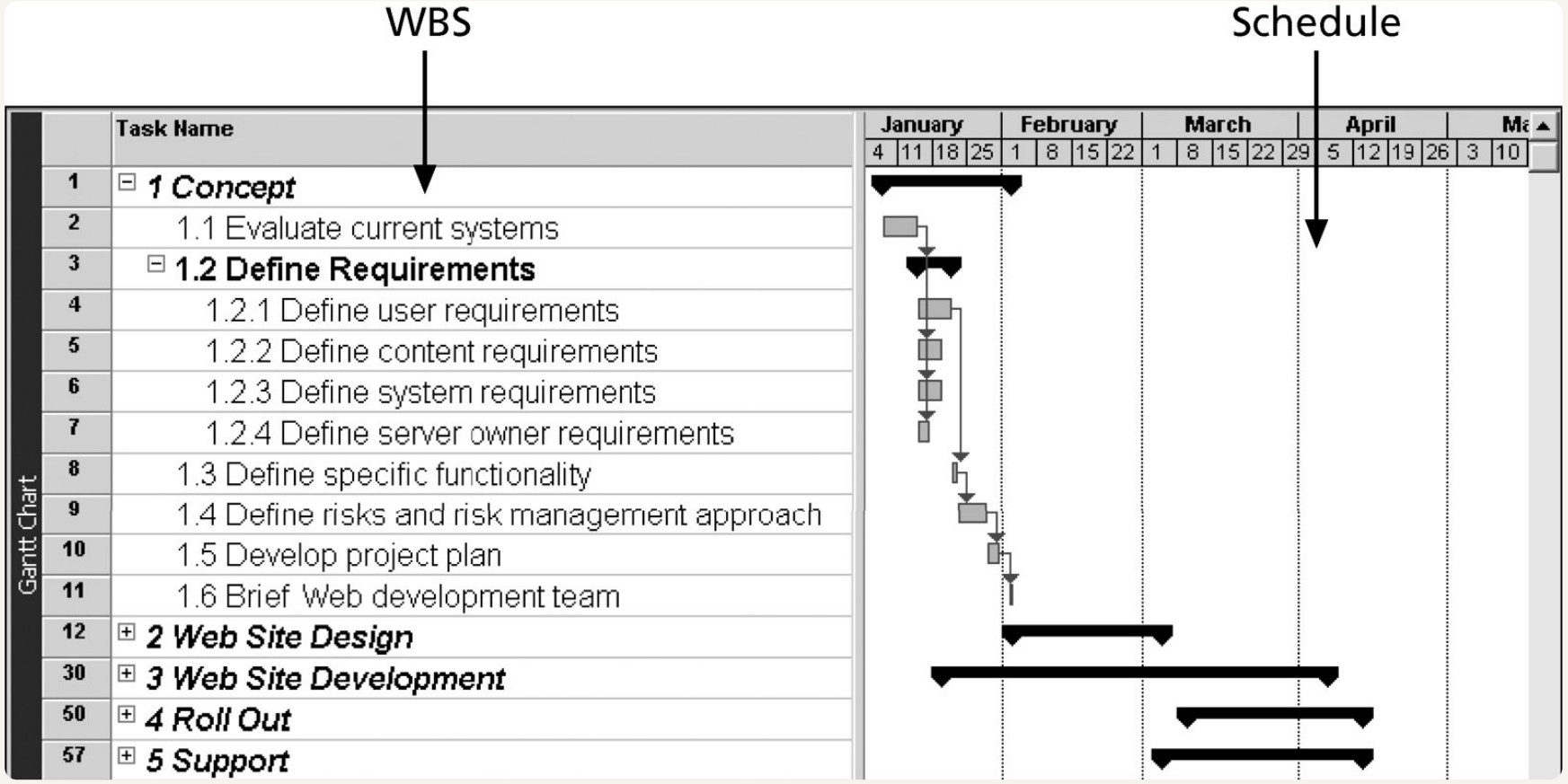
Tabular form with PMI numbering

1.1 Concept
- 1.1.1 Evaluate current systems
- 1.1.2 Define requirements
  - 1.1.2.1 Define user requirements
  - 1.1.2.2 Define content requirements
  - 1.1.2.3 Define system requirements
  - 1.1.2.4 Define server owner requirements
- 1.1.3 Define specific functionality
- 1.1.4 Define risks and risk management approach
- 1.1.5 Develop project plan
- 1.1.6 Brief Web development team

1.2 Web site design
1.3 Web site development
1.4 Roll out
1.5 Support

- Chart form shows levels 1 to 4
- Tabular form uses outline numbering
- Numbering differs between Microsoft Project and PMI styles
- Phases here: concept, web site design, web site development, roll out, support

Intranet WBS and Gantt Chart in Microsoft Project



Approaches to Developing a WBS

Guidelines and analogy

Use organization guidelines and templates, or review WBSs of similar projects and tailor them. One aircraft company used 74 previous WBSs to create a new one.

Top-down

Start with the largest items and break them into smaller items.

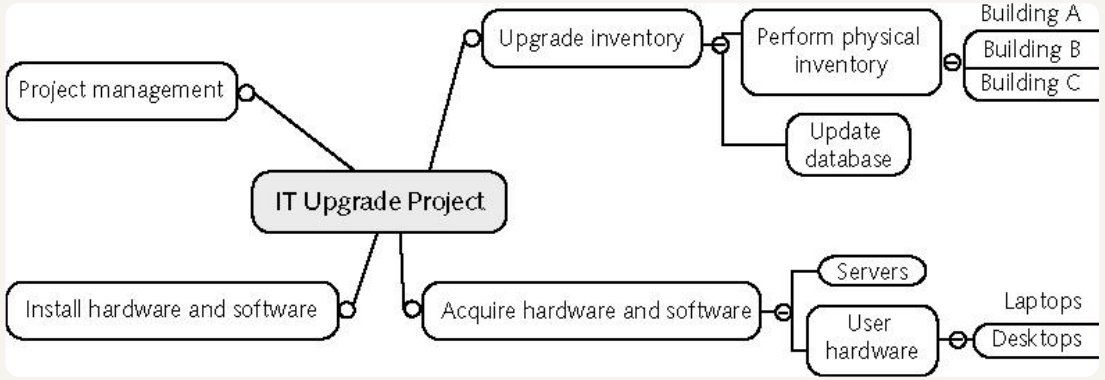
Bottom-up

Start with specific tasks and roll them up into larger items.

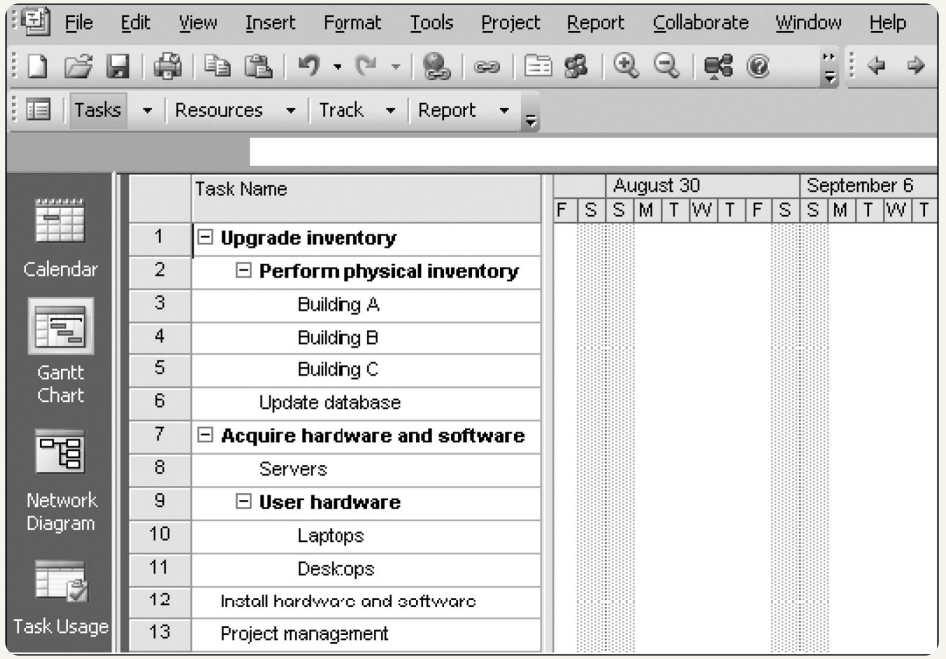
Mind-mapping

Branches radiate from a core idea to structure thoughts. Tools can export a mind map to project software.

Mind Map to WBS: IT Upgrade Project



The mind map is converted into a WBS, here shown as a Microsoft Project file.



WBS Dictionary and Scope Baseline

A WBS dictionary describes each WBS item. Scope statement + WBS + WBS dictionary = **scope baseline**, used to measure performance.

Field	Example: item 2.2 Build registration form
Description	Web form for students to enrol in courses
Responsible	[Developer name]
Deliverable / acceptance	Working form, approved in a client test
Not included	Online fee payment

Rules: a unit of work appears in one place only · each item has one responsible person · parent work = sum of its children.

Validate Scope

Formal acceptance of completed deliverables by the stakeholders.



Scope is hard to define well and harder to verify. Good configuration management and verification keep scope changes small.

Example: the client tests the registration form in a demo and signs the acceptance form for that deliverable.

Scope Problems and Best Practices

Scope creep: scope keeps growing

- Lack of user involvement
- Unneeded latest and greatest technology instead of off-the-shelf
- Failure to follow good project management processes

Best practices

- Keep scope realistic; split big projects
- Involve users (Joint Application Development)
- Use off-the-shelf hardware and software when possible
- Follow good PM processes

Control Scope

Controlling changes to the project scope. **Variance** is the difference between planned and actual performance.

Goals

- Influence the factors that cause scope changes
- Make sure changes follow integrated change control procedures
- Manage changes when they occur